Title to be chosen

2026-11-14

Introduction

Plan

- **The gap.**
  - Real-time case counts are incomplete because of reporting delays.
  - Generic forecasting models do not use the reporting structure.
  - Key references: (Höhle and Heiden 2014; Johnson et al. 2025; Reich et al. 2022).
- **Context.**
  - Nowcasting methods model reporting delays directly.
  - Time series foundation models forecast well with no training on the target data.
  - Training on domain-specific synthetic data has precedent: KernelSynth for Chronos (Ansari et al. 2024), ForecastPFN (Dooley et al. 2023), prior-fitted networks (Müller et al. 2022; Hollmann et al. 2025), and Mantis for outbreaks (Dudley et al. 2026).
  - Key references: (McGough et al. 2020; Günther et al. 2021; Wolffram et al. 2023; Ansari et al. 2024; Das, Kong, et al. 2024; Woo et al. 2024; Rasul et al. 2023; Garza, Challu, and Mergenthaler-Canseco 2023; Aksu et al. 2024; Shchur et al. 2025; Kalahasti et al. 2026; Dudley et al. 2025).
- **What we did.**
  - We fine-tuned a foundation model on synthetic outbreak data, following that precedent but fine-tuning a general model rather than training from scratch.
  - We evaluated it retrospectively and prospectively.
  - Key references: (Ansari et al. 2025; Johnson et al. 2025).

Methods

Data

Plan

- The German COVID-19 hospitalisation nowcast hub.
- The Ebola forecasting hub target and its releases.
- Key references: (Wolffram et al. 2023; KIT Metrics Lab et al. 2025; InsightNet-US 2026; MacNeil et al. 2010).

Baselines

Plan

- `randomwalk`, a random walk with no delay adjustment, and `randomwalknowcast`, a random walk nowcast.
- Both in full in the SI.
- Key references: (Johnson et al. 2025).

Foundation model setup and fine-tuning

Plan

- Chronos-2, given the reporting triangle as series: one series per place and reporting delay, each missing until reported. Introduced alongside the reporting triangle, with Figure 1B.
- Fine-tuned on synthetic data on a laptop, with checkpoints at doublings of simulated outbreaks from 1,000 upwards, each outbreak giving several forecast origins and 10 observation draws, continuing while held-out loss keeps falling. The last is the main model.
- Compared with Chronos-2 given only complete data points, to show what the ragged structure and fine-tuning add.
- Figure 1, a multi-panel schematic of our pipeline: (A) the assumptions in the epinowcast simulations, (B) the reporting triangle laid out as ragged series for Chronos-2, (C) training blocks and LoRA fine-tuning, and why it is fast, and (D) forecasting with the fine-tuned model.
- Key references: (Ansari et al. 2025, 2024; Hu et al. 2021; Das, Faw, et al. 2024).

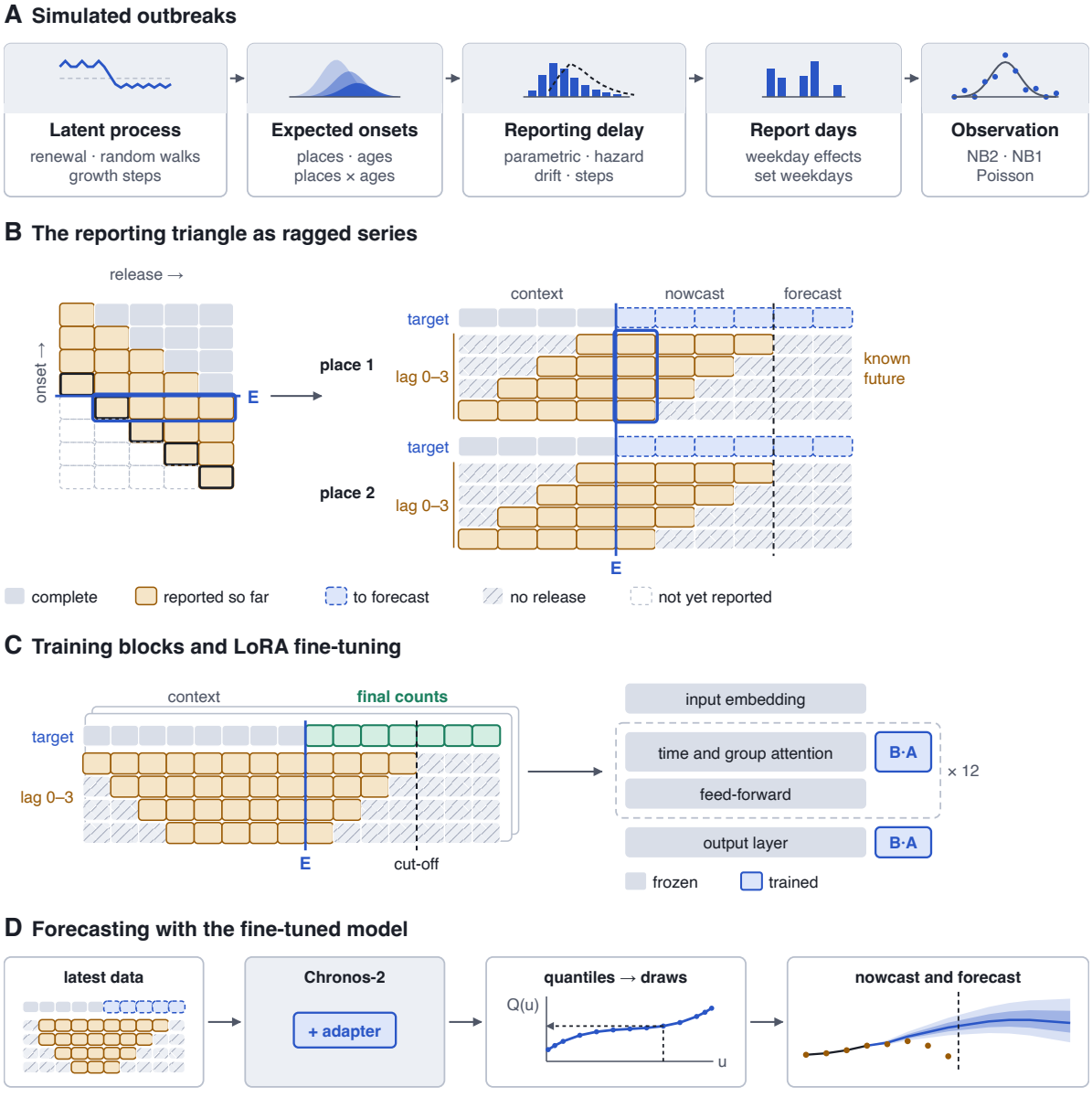


Figure 1: The fine-tuning pipeline. (A) The assumptions in the epinowcast simulations: a latent process gives expected onsets by stratum, which a reporting delay that may drift or step, report-day effects and count noise turn into a reporting triangle. (B) The reporting triangle laid out as ragged series for Chronos-2: each place has a target complete up to E and one series per lag, missing until reported (daily lag series fill only on release days), and counts reported after E enter as known-future inputs. (C) Training blocks and LoRA fine-tuning, and why it is fast: blocks cut from simulated release histories at random cut-offs are scored against the final counts, and only rank-8 adapters, about 1% of the weights, are trained. (D) Forecasting with the fine-tuned model: one forward pass gives quantiles for each step, and each draw reads them at a uniform level u and is at least the count reported so far.

Synthetic data generation

> **Plan**
>
> - Simulated with epinowcast over a grid of outbreak, delay and noise settings.
> - A table in the main text gives the grid of synthetic scenarios.
> - The full process and an overview of the epinowcast model used are in the SI.
> - Key references: (Abbott et al. 2026; Lison et al. 2024; Park et al. 2024; Ansari et al. 2024; Dooley et al. 2023; Müller et al. 2022; Hollmann et al. 2025).

Table 1: The simulation grid (r2-cal5). Ranges are chosen to be plausible across infectious diseases rather than for one disease. The reason for each is in the SI.

Setting	Range or mix
Strata	one 30%, 2-26 places 40%, 2-8 age groups 20%, 6-16 places by 3-7 ages 10%
Quiet strata	up to 90% of places seeded with almost no cases
Generative process	renewal, renewal with incubation, growth walk, log-incidence walk, growth steps, smooth growth; a sixth each
Generation time	gamma, mean 2-20 days, CV 0.3-0.7
Starting log Rt	normal, mean 0.1, sd 0.2
Starting daily growth (no generation time)	normal, mean 0.01, sd 0.02; limits plus or minus 0.1
Trajectory	turning over, flat, rising then falling, second wave; a quarter each
Seed	0.2-200 a day, log-uniform
Size	10 to 300,000 cases times strata/26; draws expected over 1.5 times the cap rejected before simulating
Duration	16-40 weeks (16-24 for places by ages)
Delay distribution	lognormal, gamma (CV 0.7 or more), exponential, or a non-parametric hazard alone
Delay mean	2-21 days, log-uniform
Longest delay	3-12 weeks, 5-10 times the delay mean
Delay changes	stable 20%; otherwise drift, 0-3 step changes, spread changes, weekly walk, week hazard effects
Report-date effects	day of week 70%, holidays 30%, report-week effects 40%, weekday-only reporting 50%, weekday switch 30%

Setting	Range or mix
Observation model	negative binomial NB2 40%, NB1 40% (dispersion matched to NB2, epinowcast #937), Poisson 20%
Overdispersion	sqrt(phi) 0.05-1.5

Correlated trajectories

i Plan

- Models that give only per-step forecasts get correlated trajectories from a Gaussian copula, with correlations estimated over a rolling window of recent forecast errors (#221).
- The same layer serves `randomwalknowcast`.

Evaluation on held-out synthetic data

i Plan

- A held-out set of simulations was not used during development.
- Sample forecasts scored with CRPS and log-scale CRPS, bias, and coverage, relative to `randomwalknowcast`.
- Scored against a rolling evaluation set of the data as reported D days later, with D each outbreak's 98% delay.
- Key references: (Bracher et al. 2021; Bosse et al. 2023; Gneiting and Raftery 2007; Gneiting, Balabdaoui, and Raftery 2007; Sherratt et al. 2023).

Case study 1: German nowcast hub

i Plan

- Compared with the hub's models, relative to the hub's own baseline, and evaluated as the hub did.
- Quantile forecasts scored with WIS and log-scale WIS, bias, and coverage, against a rolling evaluation set of the data as reported D days later (Wolffram et al. 2023).
- Daily and weekly versions may not both make sense here.
- The nowcast dates are split in half. We developed our setup on the first half only,

and the second half is held out for the results.
- Key references: (Wolffram et al. 2023; Johnson et al. 2025; Amaral et al. 2025).

Case study 2: prospective submission to the Ebola forecasting hub

i Plan

- Compared with the other submissions, relative to the hub's own baseline.
- Scored as for the held-out synthetic data, with D the empirical 98% point of the hub data's reporting delay.
- Our models' forecasts are saved prospectively, alongside the main submissions as they were made.
- Key references: (InsightNet-US 2026; Viboud et al. 2018; Funk et al. 2019).

Implementation

i Plan

- The tools we used, each cited: Python, PyTorch on the laptop GPU, chronos-forecasting, PEFT for LoRA, R and epinowcast for simulation, and scoringrules for scoring, checked against scoringutils.
- The evaluation runs as a Snakemake workflow, so every result can be reproduced from the raw data.

Results

i Plan

Each section has a figure panel. Each panel shows:

- nowcasts over time (ribbons);
- scores relative to the baseline (`randomwalknowcast` for synthetic data, the hub's baseline for each case study), with natural- and log-scale WIS (or CRPS, MAE where they make sense);
- bias;
- 50% and 90% coverage;
- score variation by location and horizon, supported by SI plots.
- for Germany, a dotted line marking the split between the development and held-out

halves, with scores split by half.

The plots in primarycensored-paper are a model for these. Each section has two paragraphs, one on visual performance and one summarising scores.
Key references: the Methods references, with (Bracher et al. 2021; Bosse et al. 2023) for the natural- and log-scale panels.

Held-out synthetic data

Plan

- Visual performance of the candidate models against `randomwalk` and `randomwalknowcast`.
- A summary of score performance.
- How scores change with the amount of fine-tuning data (doublings from 1,000 outbreaks), as an SI figure.

Placeholder summary from the workflow: the best model was epinowcast, with a relative score of 0.61 and 0.68 on the log scale, against randomwalknowcast.

Table 2: Held-out synthetic data: mean scores, coverage and number of forecasts of each model, with scores relative to randomwalknowcast. Illustrative, dummy data

study	model	metric	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	epinowcast	crps	12.6	0.12	0.143	0.522	0.876	1051	0.613	0.681
synthetic	fm_chronos	crps	daily_complete	22.3	0.211	-0.227	0.426	0.821	1051	1.08	1.2
synthetic	fm_chronos	crps	daily_ragged	23.4	0.187	0.369	0.335	0.676	1051	1.14	1.07
synthetic	fm_chronos	crps	daily_ragged_ft	13.3	0.129	0.105	0.495	0.882	1051	0.648	0.734
synthetic	fm_chronos	crps	weekly_complete	24	0.216	-0.271	0.426	0.817	1051	1.17	1.23
synthetic	fm_chronos	crps	weekly_ragged	19.9	0.181	-0.0368	0.435	0.793	1051	0.969	1.03
synthetic	fm_chronos	crps	weekly_ragged_ft	17.4	0.14	0.197	0.538	0.882	1051	0.847	0.796
synthetic	randomwalk	crps		22.3	0.186	-0.163	0.574	0.95	1051	1.09	1.06

study	model	metric	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
synthetic	randomwalknowcast	crps	20.6	0.176	0.0841	0.656	0.967	1051	1	1

German nowcast hub

> **Plan**
>
> - Visual performance against the other models.
> - A summary of score performance.
> - Key references: (Wolffram et al. 2023; Amaral et al. 2025).

Placeholder summary from the workflow: the best model was MeanEnsemble, with a relative score of 0.11 and 0.14 on the log scale, against KIT-frozen_baseline.

Table 3: German nowcast hub: mean scores, coverage and number of forecasts of each model, with scores relative to KIT-frozen_baseline.

study	model	metric	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	Epiforecasts	wis	22.6	0.0302	0.0364	0.375	0.774	9185	0.137	0.159
germany	ILM	wis	78	0.0404	0.677	0.281	0.637	2716	0.224	0.22
germany	KIT	wis	24.5	0.032	0.00999	0.526	0.929	9200	0.148	0.168
germany	KIT-frozen_baseline	wis	165	0.191	-0.941	0.0373	0.0373	9200	1	1
germany	LMU	wis	26.4	0.0339	0.00453	0.269	0.61	8495	0.148	0.171
germany	MeanEnsemble	wis	18	0.0271	0.174	0.419	0.828	9177	0.11	0.143

study	model	metric	score	log_score	bias	coverage_50	coverage_95	n	rel_score	rel_log_score
germany	Media-a-nEnsemble	wis	18.4	0.0271	0.11	0.422	0.83	9177	0.112	0.143
germany	RIVM	wis	28	0.0376	0.397	0.233	0.553	9200	0.169	0.197
germany	RKI	wis	30.1	0.0489	0.215	0.163	0.304	6800	0.202	0.255
germany	SU	wis	26.3	0.0341	0.0796	0.376	0.781	9200	0.159	0.179
germany	SZ	wis	44.2	0.0538	-0.196	0.149	0.37	9131	0.266	0.28
germany	randomwalk	wis	98.1	0.0996	-0.42	0.212	0.467	9200	0.593	0.522
germany	randomwal-knowcast	wis	22.6	0.03	-0.137	0.525	0.901	9200	0.137	0.157

Ebola hub submissions

> **Plan**
>
> - Visual performance against the other submissions.
> - A summary of score performance.

Placeholder summary from the workflow: the best model was randomwalknowcast, with a relative score of 0.73 and 0.85 on the log scale, against BDBVHub-baseline.

Table 4: Ebola hub submissions: mean scores, coverage and number of forecasts of each model, with scores relative to BDBVHub-baseline. Illustrative, dummy data

study	model	metric	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	BDBVHub-baseline	crps	4.24	0.293	-0.00217	0.711	0.964	810	1	1
ebola	fm_chronos_complete	crps_weekly_com	5.63	-0.347	-0.102	0.562	0.886	810	1.33	1.18

study	model	metric	score	log_score	bias	coverage_50	coverage_90	n	rel_score	rel_log_score
ebola	fm_chronos	crps_weekly_ragged	3.63	0.293	-0.0367	0.533	0.83	810	0.856	1
ebola	fm_chronos	crps_weekly_ragged	3.2	0.237	0.116	0.631	0.931	810	0.755	0.809
ebola	hub_ensemble	crps	3.26	0.24	0.14	0.599	0.904	810	0.77	0.817
ebola	hub_team_a	crps	5.28	0.326	0.38	0.432	0.777	810	1.25	1.11
ebola	hub_team_b	crps	3.53	0.294	0.00569	0.662	0.964	810	0.833	1
ebola	random_walk	crps	4.06	0.308	-0.0843	0.673	0.965	810	0.958	1.05
ebola	random_walk_nowcast	crps	3.11	0.25	0.0655	0.737	0.983	810	0.734	0.853

Discussion

Plan

- **Summary** of what we did.
- **Strengths and weaknesses.**
  - Weaknesses: only one foundation model, only one source of synthetic data, and limited case studies.
  - Weakness: we did not compare against epinowcast, or, for the synthetic study, against the data-generating model itself.
  - Strengths: both retrospective and real-time case studies, and synthetic data that are easy to tune to a setting.
  - Strength: fine-tuning ran locally on consumer hardware (a laptop), so it is accessible.
  - Weakness: fine-tuning used the laptop's GPU, which made it several times faster, and not everyone has one. On a laptop CPU alone it would still run in hours.
  - Key references: (Kalahasti et al. 2026; Meyer et al. 2025).
- **Domain knowledge through synthetic data.**
  - Domain knowledge enters through the synthetic data source.
  - Fingers crossed it improves results; to fill in from the results.
  - Key references: (Dudley et al. 2026, 2025; Dooley et al. 2023; Hollmann et al.

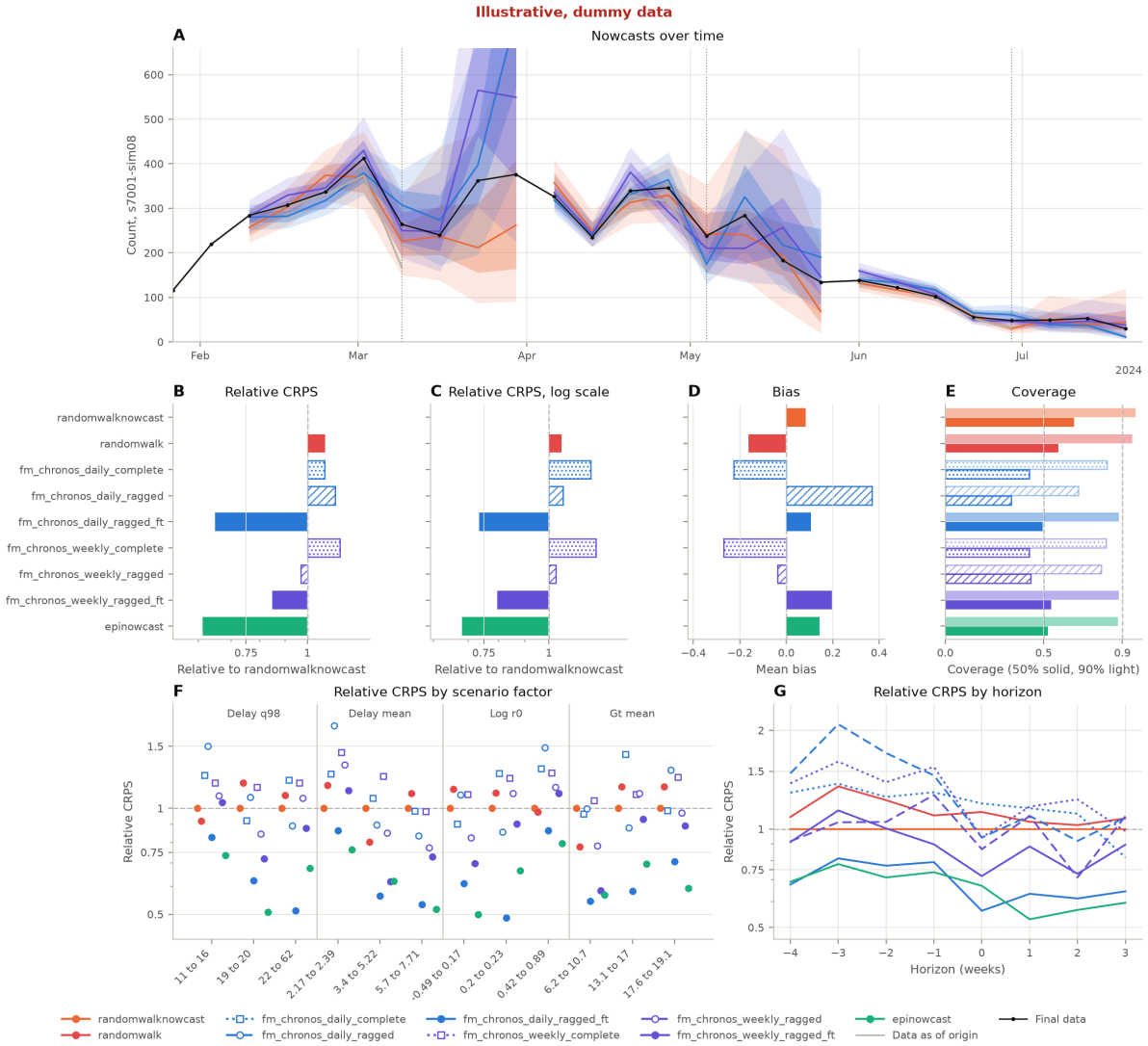


Figure 2: Held-out synthetic data. (A) Nowcasts over time; (B, C) CRPS relative to `randomwalknowcast` on the natural and log scales; (D) bias; (E) 50% and 90% coverage; (F) relative CRPS by scenario factor; (G) relative CRPS by horizon. Illustrative, dummy data

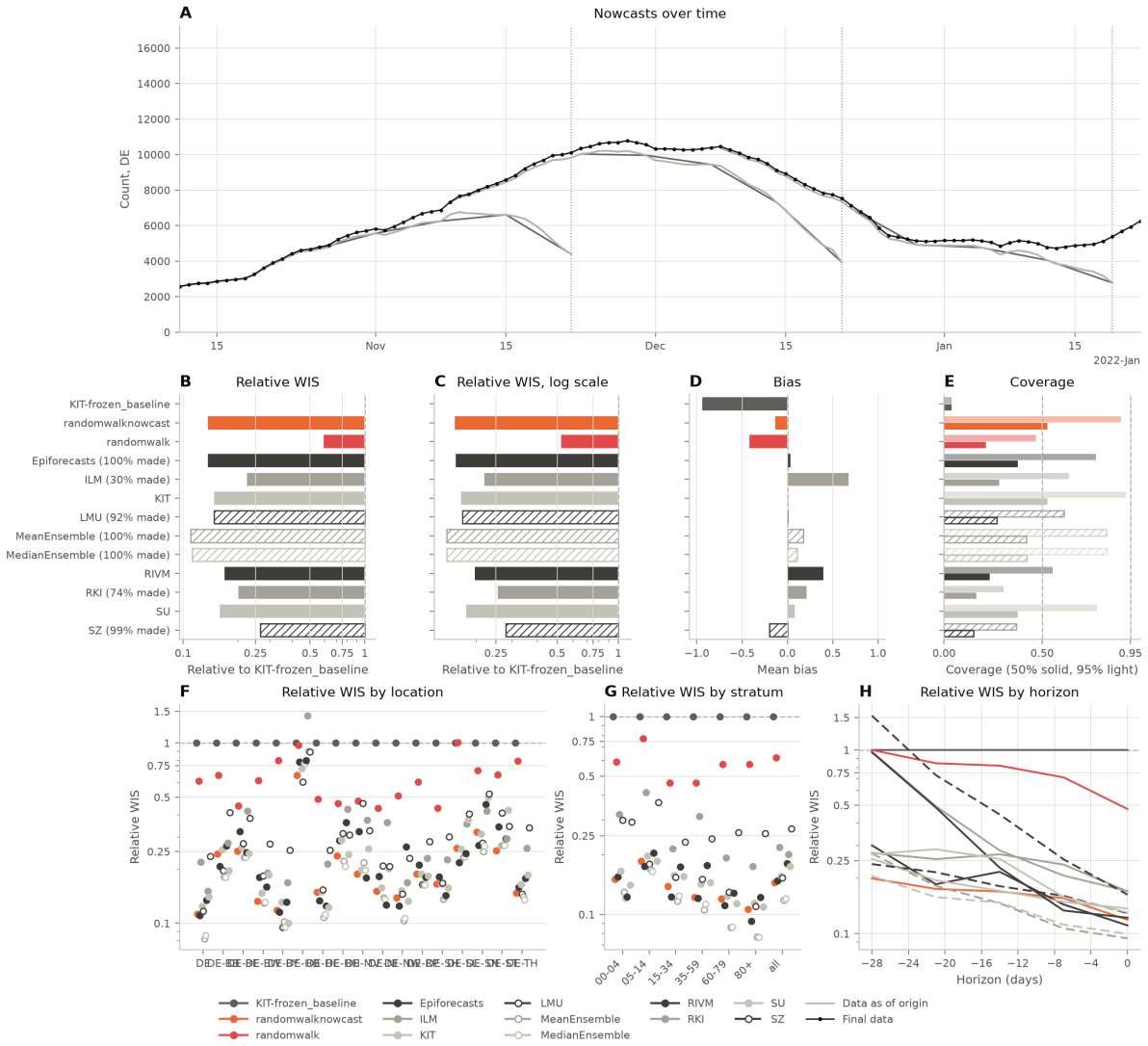


Figure 3: German nowcast hub. (A) Nowcasts over time, with a dotted line at the split between the development and held-out halves; (B, C) WIS relative to `KIT-frozen_baseline` on the natural and log scales; (D) bias; (E) 50% and 95% coverage; (F) relative WIS by location; (G) by age group; (H) by horizon.

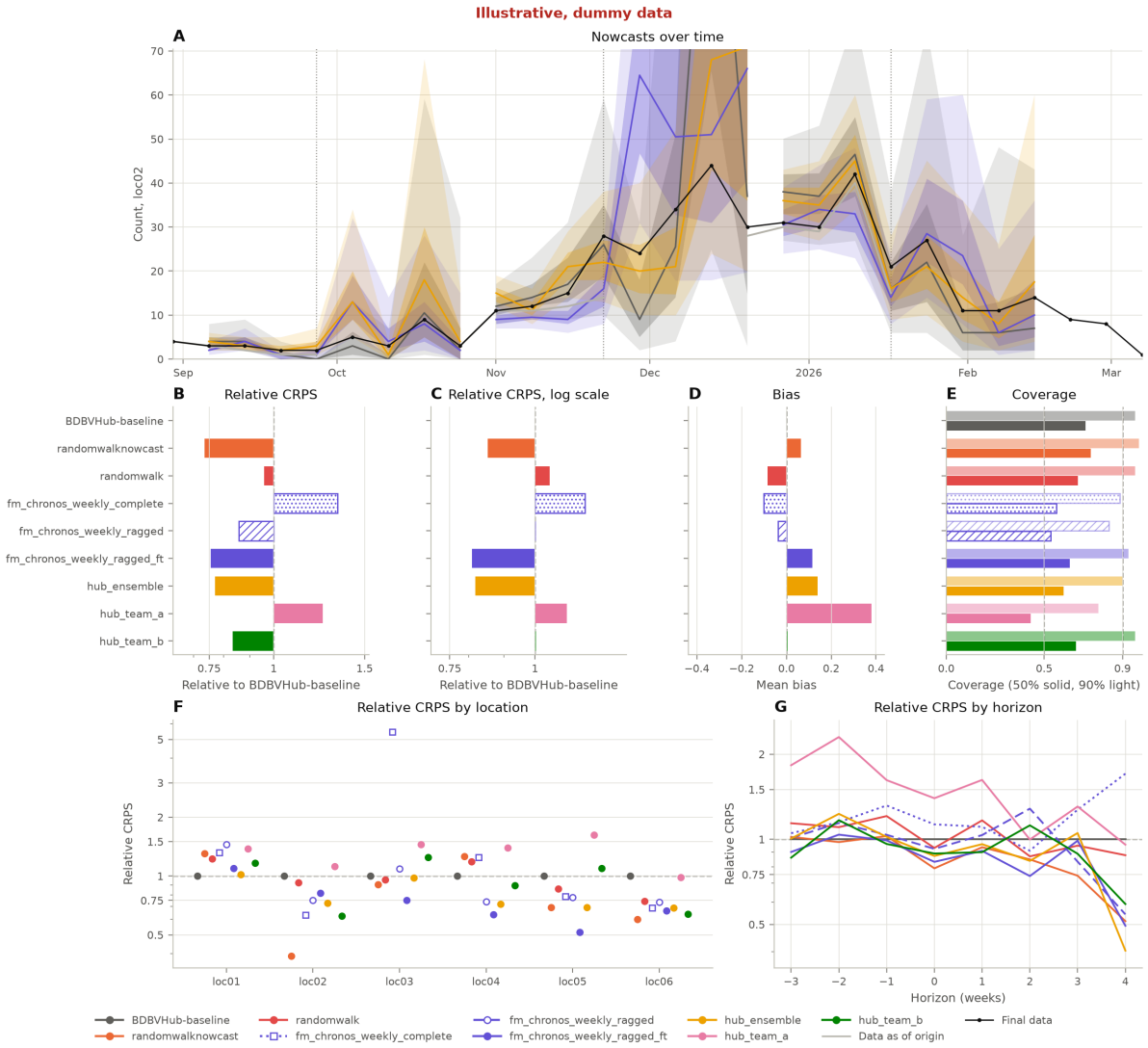


Figure 4: Ebola hub submissions. (A) Nowcasts over time; (B, C) CRPS relative to `BDBVHub-baseline` on the natural and log scales; (D) bias; (E) 50% and 90% coverage; (F) relative CRPS by location; (G) by horizon. Illustrative, dummy data

2025; Ansari et al. 2025).

- **Against the literature.**
  - Other uses of foundation models in infectious disease modelling. Key references: (Kalahasti et al. 2026; Cai, Huang, and Xu 2026; Wang et al. 2025; Jafari et al. 2026; Du et al. 2025; Li et al. 2025; Aygün et al. 2025; Martinson et al. 2026).
  - Fine-tuning in other domains. Key references: (Laglil et al. 2026; Das, Faw, et al. 2024; Hu et al. 2021).
  - Other nowcasting approaches. Key references: (Bastos et al. 2019; Seaman et al. 2022; Bergström et al. 2022; Sumalinab et al. 2024; Schneble et al. 2021; Kassteele, Eilers, and Wallinga 2019; Overton et al. 2023; Mellor et al. 2025; Tang et al. 2025; Stoner, Economou, and Halliday 2020; Kamarthi, Rodríguez, and Prakash 2021; Koemen et al. 2025; Wilsens, Antonio, and Claeskens 2025).
- **A possible emulator.**
  - Fine-tuning on epinowcast simulations may give a fast emulator of that package.
  - We did not compare against epinowcast, so we cannot say how close it is.
  - This could be useful for other slow, long-running packages.
- **Further work.**
  - Other foundation models.
  - More case studies.
  - A comparison with custom neural networks.
  - Testing the fine-tuned model as an emulator of epinowcast.
  - Fine-tuning with other epidemiological features as covariates, such as distances between locations or mobility.
  - Key references: (Das, Kong, et al. 2024; Woo et al. 2024; Rasul et al. 2023; Kamarthi, Rodríguez, and Prakash 2021; Koemen et al. 2025; KIT Metrics Lab, Bracher, and Schienle 2025).
- **Conclusions.**
  - Reuse state-of-the-art generic forecasting models with domain-specific insight.
  - Fine-tuning on a package's simulations could give a fast emulator of that package, which may help other long-running packages.
  - It shows promise, and more work is needed.

Funding

Acknowledgements

Data and software availability

AI disclosure

References

Abbott, Sam, Adrian Lison, Sebastian Funk, Carl Pearson, Hugo Gruson, Felix Guenther, Michael DeWitt, James Mba Azam, and Jessalyn Sebastian. 2026. *Epinowcast: A Bayesian Framework for Real-Time Infectious Disease Surveillance*. https://doi.org/10.5281/zenodo.5637165.

Aksu, Taha, Gerald Woo, Juncheng Liu, Xu Liu, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. 2024. "GIFT-Eval: A Benchmark for General Time Series Forecasting Model Evaluation." arXiv. https://doi.org/10.48550/arXiv.2410.10393.

Amaral, André Victor Ribeiro, Daniel Wolffram, Paula Moraga, and Johannes Bracher. 2025. "Post-Processing and Weighted Combination of Infectious Disease Nowcasts." *PLoS Computational Biology* 21 (3): e1012836. https://doi.org/10.1371/journal.pcbi.1012836.

Ansari, Abdul Fatir, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, et al. 2025. "Chronos-2: From Univariate to Universal Forecasting." arXiv. https://doi.org/10.48550/arXiv.2510.15821.

Ansari, Abdul Fatir, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, et al. 2024. "Chronos: Learning the Language of Time Series." *Transactions on Machine Learning Research*. https://doi.org/10.48550/arXiv.2403.07815.

Aygün, Eser, Anastasiya Belyaeva, Gheorghe Comanici, Marc Coram, Hao Cui, Jake Garrison, Renee Johnston Anton Kast, et al. 2025. "An AI System to Help Scientists Write Expert-Level Empirical Software." arXiv. https://doi.org/10.48550/arXiv.2509.06503.

Bastos, Leonardo S, Theodoros Economou, Marcelo F C Gomes, Daniel A M Villela, Flavio C Coelho, Oswaldo G Cruz, Oliver Stoner, Trevor Bailey, and Claudia T Codeço. 2019. "A Modelling Approach for Correcting Reporting Delays in Disease Surveillance Data." *Statistics in Medicine* 38 (22): 4363–77. https://doi.org/10.1002/sim.8303.

Bergström, Fanny, Felix Günther, Michael Höhle, and Tom Britton. 2022. "Bayesian Nowcasting with Leading Indicators Applied to COVID-19 Fatalities in Sweden." *PLOS Computational Biology* 18 (12): e1010767. https://doi.org/10.1371/journal.pcbi.1010767.

Bosse, Nikos I., Sam Abbott, Anne Cori, Edwin van Leeuwen, Johannes Bracher, and Sebastian Funk. 2023. "Scoring Epidemiological Forecasts on Transformed Scales." *PLOS Computational Biology* 19 (8): e1011393. https://doi.org/10.1371/journal.pcbi.1011393.

Bracher, Johannes, Evan L. Ray, Tilmann Gneiting, and Nicholas G. Reich. 2021. "Evaluating Epidemic Forecasts in an Interval Format." *PLOS Computational Biology* 17 (2): e1008618. https://doi.org/10.1371/journal.pcbi.1008618.

Cai, Peng, Guoyu Huang, and Ximing Xu. 2026. "Benchmarking Time Series Foundation Models for Multi-Country Infectious Disease Forecasting: A Retrospective Evaluation Across over 70 Diseases." *BMC Public Health*, October. https://doi.org/10.1186/s12889-026-29918-1.

Das, Abhimanyu, Matthew Faw, Rajat Sen, and Yichen Zhou. 2024. "In-Context Fine-Tuning for Time-Series Foundation Models." arXiv. https://doi.org/10.48550/arXiv.2410.24087.

Das, Abhimanyu, Weihao Kong, Rajat Sen, and Yichen Zhou. 2024. "A Decoder-Only Foundation Model for Time-Series Forecasting." In *Proceedings of the 41st International Conference on Machine Learning*. https://doi.org/10.48550/arXiv.2310.10688.

Dooley, Samuel, Gurnoor Singh Khurana, Chirag Mohapatra, Siddartha Naidu, and Colin White. 2023. "ForecastPFN: Synthetically-Trained Zero-Shot Forecasting." In *Advances in Neural Information Processing Systems*. https://doi.org/10.48550/arXiv.2311.01933.

Du, Hongru, Yang Zhao, Jianan Zhao, Shaochong Xu, Xihong Lin, Yiran Chen, Lauren M. Gardner, and Hao Frank Yang. 2025. "Advancing Real-Time Infectious Disease Forecasting Using Large Language Models." *Nature Computational Science* 5 (6): 467–80. https://doi.org/10.1038/s43588-025-00798-6.

Dudley, Carson, Reiden Magdaleno, Christopher Harding, and Marisa Eisenberg. 2025. "Simulation as Supervision: Mechanistic Pretraining for Scientific Discovery." arXiv. https://doi.org/10.48550/arXiv.2507.08977.

Dudley, Carson, Reiden Magdaleno, Christopher Harding, Ananya Sharma, Emily Martin, and Marisa Eisenberg. 2026. "Mantis: A Foundation Model for Mechanistic Disease Forecasting." *Proceedings of the National Academy of Sciences* 123 (40). https://doi.org/10.1073/pnas.2602542123.

Funk, Sebastian, Anton Camacho, Adam J. Kucharski, Rachel Lowe, Rosalind M. Eggo, and W. John Edmunds. 2019. "Assessing the Performance of Real-Time Epidemic Forecasts: A Case Study of Ebola in the Western Area Region of Sierra Leone, 2014-15." *PLOS Computational Biology* 15 (2): e1006785. https://doi.org/10.1371/journal.pcbi.1006785.

Garza, Azul, Cristian Challu, and Max Mergenthaler-Canseco. 2023. "TimeGPT-1." arXiv. https://doi.org/10.48550/arXiv.2310.03589.

Gneiting, Tilmann, Fadoua Balabdaoui, and Adrian E. Raftery. 2007. "Probabilistic Forecasts, Calibration and Sharpness." *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 69 (2): 243–68. https://doi.org/10.1111/j.1467-9868.2007.00587.x.

Gneiting, Tilmann, and Adrian E Raftery. 2007. "Strictly Proper Scoring Rules, Prediction, and Estimation." *Journal of the American Statistical Association* 102 (477): 359–78. https://doi.org/10.1198/016214506000001437.

Günther, Felix, Andreas Bender, Katharina Katz, Helmut Küchenhoff, and Michael Höhle. 2021. "Nowcasting the COVID-19 Pandemic in Bavaria." *Biom. J.* 63 (3): 490–502. https://doi.org/10.1002/bimj.202000112.

Höhle, Michael, and Matthias an der Heiden. 2014. "Bayesian Nowcasting During the STEC O104:H4 Outbreak in Germany, 2011." *Biometrics* 70 (4): 993–1002. https://doi.org/10.1111/biom.12194.

Hollmann, Noah, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. 2025. "Accurate Predictions

on Small Data with a Tabular Foundation Model." *Nature* 637 (8045): 319–26. https://doi.org/10.1038/s41586-024-08328-6.

Hu, Edward J., Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. "LoRA: Low-Rank Adaptation of Large Language Models." arXiv. https://doi.org/10.48550/arXiv.2106.09685.

InsightNet-US. 2026. "BDBV Modeling Hub: Modeling Hub for Bundibugyo Ebola." GitHub. https://github.com/InsightNet-US/BDBV-Modeling-Hub.

Jafari, Alireza, Judy Fox, Geoffrey C. Fox, Madhav Marathe, and Aniruddha Adiga. 2026. "Understanding Key Features of Time Series Foundation Models from Epidemic Forecasting." arXiv. https://doi.org/10.48550/arXiv.2606.19560.

Johnson, Kaitlyn E, Maria L Tang, Emily Tyszka, Laura Jones, Barbora Nemcova, Daniel Wolffram, Rosa Ergas, et al. 2025. "Baseline Nowcasting Methods for Handling Delays in Epidemiological Data." *Wellcome Open Research* 10: 614. https://doi.org/10.12688/wellcomeopenres.25027.2.

Kalahasti, Suprabhath, Benjamin Faucher, Boxuan Wang, Claudio Ascione, Ricardo Carbajal, Maxime Enault, Christophe Vincent Cassis, et al. 2026. "Foundation Models for Time Series Forecasting and Policy Evaluation in Infectious Disease Epidemics." *Epidemics* 55 (June): 100916. https://doi.org/10.1016/j.epidem.2026.100916.

Kamarthi, Harshavardhan, Alexander Rodríguez, and B. Aditya Prakash. 2021. "Back2Future: Leveraging Backfill Dynamics for Improving Real-Time Predictions in Future." In *Advances in Neural Information Processing Systems*. https://doi.org/10.48550/arXiv.2106.04420.

Kassteele, Jan van de, Paul H C Eilers, and Jacco Wallinga. 2019. "Nowcasting the Number of New Symptomatic Cases During Infectious Disease Outbreaks Using Constrained P-Spline Smoothing." *Epidemiology* 30 (5): 737–45. https://doi.org/10.1097/EDE.0000000000001050.

KIT Metrics Lab, Johannes Bracher, and Melanie Schienle. 2025. "RESPINOW-Hub: Nowcasting and Short-term Forecasting Platform for Respiratory Diseases." Karlsruhe Institute of Technology; GitHub. https://github.com/KITmetricslab/RESPINOW-Hub.

KIT Metrics Lab, Johannes Bracher, Daniel Wolffram, and Melanie Schienle. 2025. "Hospitalization Nowcast Hub: Collecting Nowcasts of the 7-Day Hospitalization Incidence in Germany." Karlsruhe Institute of Technology; GitHub. https://github.com/KITmetricslab/hospitalization-nowcast-hub.

Koemen, Silas, Nuno R. Faria, Leonardo S. Bastos, Oliver Ratmann, and André Victor Ribeiro Amaral. 2025. "Fast and Trustworthy Nowcasting of Dengue Fever: A Case Study Using Attention-Based Probabilistic Neural Networks in São Paulo, Brazil." medRxiv. https://doi.org/10.1101/2025.05.05.25326971.

Laglil, Morad, Bertrand Pracca, Emilie Devijver, and Eric Gaussier. 2026. "Foundation Models and Fine-Tuning: Toward a New Generation of Models for Time Series Forecasting." arXiv. https://doi.org/10.48550/arXiv.2607.23146.

Li, Chenxiang, Qiqiao Zhang, Yue Zhang, Bowen Zhao, Jule Yang, Li Qi, Jun Ding, and Dechao Tian. 2025. "Fine-Tuned Large Language Models Enhance Influenza Forecasting." medRxiv. https://doi.org/10.1101/2025.03.27.25324747.

Lison, Adrian, Sam Abbott, Jana Huisman, and Tanja Stadler. 2024. "Generative Bayesian

Modeling to Nowcast the Effective Reproduction Number from Line List Data with Missing Symptom Onset Dates." Edited by Tom Britton. *PLOS Computational Biology* 20 (4): e1012021. https://doi.org/10.1371/journal.pcbi.1012021.

MacNeil, Adam, Eileen C. Farnon, Joseph Wamala, Sam Okware, Deborah L. Cannon, Zachary Reed, Jonathan S. Towner, et al. 2010. "Proportion of Deaths and Clinical Features in Bundibugyo Ebola Virus Infection, Uganda." *Emerging Infectious Diseases* 16 (12): 1969–72. https://doi.org/10.3201/eid1612.100627.

Martinson, Sarah, Michael P. Brenner, Martyna Plomecka, Brian P. Williams, Nicholas G. Reich, and Zahra Shamsi. 2026. "Prospective Multi-Pathogen Disease Forecasting Using Autonomous LLM-Guided Tree Search." arXiv. https://doi.org/10.48550/arXiv.2605.16238.

McGough, Sarah F., Michael A. Johansson, Marc Lipsitch, and Nicolas A. Menzies. 2020. "Nowcasting by Bayesian Smoothing: A Flexible, Generalizable Model for Real-Time Epidemic Tracking." *PLoS Computational Biology* 16: e1007735. https://doi.org/10.1371/journal.pcbi.1007735.

Mellor, Jonathon, Maria L Tang, Emilie Finch, Rachel Christie, Oliver Polhill, Christopher E Overton, Ann Hoban, Amy Douglas, Sarah R Deeny, and Thomas Ward. 2025. "An Application of Nowcasting Methods: Cases of Norovirus During the Winter 2023/2024 in England." *PLOS Computational Biology* 21 (2): e1012849. https://doi.org/10.1371/journal.pcbi.1012849.

Meyer, Marcel, Sascha Kaltenpoth, Kevin Zalipski, and Oliver Müller. 2025. "Rethinking Evaluation in the Era of Time Series Foundation Models: (Un)known Information Leakage Challenges." arXiv. https://doi.org/10.48550/arXiv.2510.13654.

Müller, Samuel, Noah Hollmann, Sebastian Pineda Arango, Josif Grabocka, and Frank Hutter. 2022. "Transformers Can Do Bayesian Inference." In *International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2112.10510.

Overton, Christopher E, Sam Abbott, Rachel Christie, Fergus Cumming, Julie Day, Owen Jones, Rob Paton, Charlie Turner, and Thomas Ward. 2023. "Nowcasting the 2022 Mpox Outbreak in England." *PLoS Comput. Biol.* 19 (9): e1011463. https://doi.org/10.1371/journal.pcbi.1011463.

Park, Sang Woo, Andrei R. Akhmetzhanov, Kelly Charniga, Anne Cori, Nicholas G. Davies, Jonathan Dushoff, Sebastian Funk, et al. 2024. "Estimating Epidemiological Delay Distributions for Infectious Diseases." medRxiv. https://doi.org/10.1101/2024.01.12.24301247.

Rasul, Kashif, Arjun Ashok, Andrew Robert Williams, Hena Ghonia, Rishika Bhagwatkar, Arian Khorasani, Mohammad Javad Darvishi Bayazi, et al. 2023. "Lag-Llama: Towards Foundation Models for Probabilistic Time Series Forecasting." arXiv. https://doi.org/10.48550/arXiv.2310.08278.

Reich, Nicholas G., Justin Lessler, Sebastian Funk, Cecile Viboud, Alessandro Vespignani, Ryan J. Tibshirani, Katriona Shea, et al. 2022. "Collaborative Hubs: Making the Most of Predictive Epidemic Modeling." *American Journal of Public Health* 112 (6): 839–42. https://doi.org/10.2105/AJPH.2022.306831.

Schneble, Marc, Giacomo De Nicola, Göran Kauermann, and Ursula Berger. 2021. "Nowcasting Fatal COVID-19 Infections on a Regional Level in Germany." *Biometrical Journal* 63 (3): 471–89. https://doi.org/10.1002/bimj.202000143.

Seaman, Shaun R, Pantelis Samartsidis, Meaghan Kall, and Daniela De Angelis. 2022. "Nowcasting COVID-19 Deaths in England by Age and Region." *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 71 (5): 1266–81. https://doi.org/10.1111/rssc.12576.

Shchur, Oleksandr, Abdul Fatir Ansari, Caner Turkmen, Lorenzo Stella, Nick Erickson, Pablo Guerron, Michael Bohlke-Schneider, and Yuyang Wang. 2025. "fev-bench: A Realistic Benchmark for Time Series Forecasting." arXiv. https://doi.org/10.48550/arXiv.2509.26468.

Sherratt, Katharine, Hugo Gruson, Rok Grah, Helen Johnson, Rene Niehus, Bastian Prasse, Frank Sandmann, et al. 2023. "Predictive Performance of Multi-Model Ensemble Forecasts of COVID-19 Across European Nations." Edited by Amy Wesolowski, Neil M Ferguson, Jeffrey L Shaman, and Sen Pei. *eLife* 12 (April): e81916. https://doi.org/10.7554/eLife.81916.

Stoner, Oliver, Theo Economou, and Alba Halliday. 2020. "A Powerful Modelling Framework for Nowcasting and Forecasting COVID-19 and Other Diseases." arXiv. https://doi.org/10.48550/arXiv.1912.05965.

Sumalinab, Bryan, Oswaldo Gressani, Niel Hens, and Christel Faes. 2024. "Bayesian Nowcasting with Laplacian-P-Splines." *Journal of Computational and Graphical Statistics* 34 (2): 718–28. https://doi.org/10.1080/10618600.2024.2395414.

Tang, Maria L, Ian S McFarlane, Christopher E Overton, Erjola Hani, Vanessa Saliba, Gareth J Hughes, Paul Crook, Thomas Ward, and Jonathon Mellor. 2025. "Nowcasting Cases and Trends During the Measles 2023/24 Outbreak in England." *Journal of Infection* 91 (3): 106569. https://doi.org/10.1016/j.jinf.2025.106569.

Viboud, Cécile, Kaiyuan Sun, Robert Gaffey, Marco Ajelli, Laura Fumanelli, Stefano Merler, Qian Zhang, Gerardo Chowell, Lone Simonsen, and Alessandro Vespignani. 2018. "The RAPIDD Ebola Forecasting Challenge: Synthesis and Lessons Learnt." *Epidemics* 22 (March): 13–21. https://doi.org/10.1016/j.epidem.2017.08.002.

Wang, Yanqin, Guoyu Huang, Ceran Chen, Qiu Li, and Ximing Xu. 2025. "A Comparative Study of Time Series Foundation Models for Hand, Foot, and Mouth Disease Forecasting: TimesFM, Moirai, and Traditional Approaches." *Frontiers in Public Health* 13 (September). https://doi.org/10.3389/fpubh.2025.1634138.

Wilsens, Paul, Katrien Antonio, and Gerda Claeskens. 2025. "Machine Learning in an Expectation-Maximisation Framework for Nowcasting." arXiv. https://doi.org/10.48550/arXiv.2512.07335.

Wolffram, Daniel, Sam Abbott, Matthias an der Heiden, Sebastian Funk, Felix Günther, Davide Hailer, Stefan Heyder, et al. 2023. "Collaborative Nowcasting of COVID-19 Hospitalization Incidences in Germany." *PLOS Computational Biology* 19 (8): 1–25. https://doi.org/10.1371/journal.pcbi.1011394.

Woo, Gerald, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. 2024. "Unified Training of Universal Time Series Forecasting Transformers." In *Proceedings of the 41st International Conference on Machine Learning*. https://doi.org/10.48550/arXiv.2402.02592.